

Azure Kubernetes Service (AKS) – Interview Questions and Answers

Basics of AKS

- Q: What is Kubernetes?

A: Kubernetes is an open-source container orchestration platform used to deploy, scale, and manage containerized applications.

- Q: What is Azure Kubernetes Service (AKS)?

A: AKS is a managed Kubernetes service by Microsoft Azure that simplifies cluster management by handling control plane operations.

- Q: Difference between Kubernetes and AKS?

A: Kubernetes is open-source, whereas AKS is a managed service on Azure.

Architecture & Core Concepts

- Q: What is a Pod?

A: A Pod is the smallest deployable unit in Kubernetes and can contain one or more containers.

- Q: What is a Node Pool?

A: A node pool is a group of worker nodes with the same configuration.

- Q: What is a Deployment?

A: Deployment ensures the desired number of pod replicas are running.

Networking

- Q: What is Ingress?

A: Ingress manages external HTTP/HTTPS traffic routing to services.

- Q: Service vs Ingress?

A: Service exposes apps, Ingress manages routing and SSL.

Storage

- Q: What is a Persistent Volume?

A: PV represents storage resources in the cluster.

- Q: What is a PVC?

A: PVC is a request for storage by a pod.

- Q: AKS Storage options?

A: Azure Disk, Azure File, Azure Blob.

Security

- Q: How is authentication handled?

A: Using Azure AD and Kubernetes RBAC.

- Q: What is RBAC?

A: Role-Based Access Control limits access to resources.

- Q: How to store secrets?

A: Kubernetes Secrets or Azure Key Vault.

Scaling & Availability

- Q: What is HPA?

A: Scales pods based on CPU or memory usage.

- Q: What is Cluster Autoscaler?

A: Automatically scales the number of nodes.

- Q: How does AKS ensure HA?

A: Multiple nodes, replicas, and availability zones.

CI/CD & DevOps

- Q: How do you deploy apps to AKS?

A: Using kubectl, Helm, Azure DevOps, or GitHub Actions.

- Q: What is Helm?

A: Helm is a package manager for Kubernetes.

- Q: What is a Helm Chart?

A: A collection of YAML files defining Kubernetes resources.

Monitoring & Troubleshooting

- Q: How do you monitor AKS?

A: Using Azure Monitor, Container Insights, Prometheus.

- Q: Pod in Pending state reasons?

A: Insufficient resources, PVC issues, or node selector mismatch.

- Q: How to check logs?

A: Using `kubectl logs <pod-name>`.

1. DevOps & Cloud-Native Experience

Q1. Explain your DevOps experience and how it aligns with cloud-native technologies.

Answer:

I have over 7+ years of experience in DevOps, primarily working with cloud-native architectures on Azure. My work involves containerizing applications using Docker, orchestrating them with Kubernetes, and deploying them on Azure Kubernetes Service (AKS). I focus heavily on CI/CD automation, Infrastructure as Code, observability, and reliability engineering.

Q2. What does cloud-native architecture mean to you?

Answer:

Cloud-native architecture involves building applications that are:

- Containerized
- Microservices-based
- Highly scalable and resilient
- Managed via orchestration platforms like Kubernetes
- Deployed using CI/CD pipelines and IaC

AKS, Docker, Helm, and Azure DevOps form the backbone of this approach.

2. Kubernetes (Core & Advanced)

Q3. How do you manage Kubernetes clusters in production?

Answer:

In production, I manage Kubernetes clusters by:

- Using separate namespaces for environments
 - Implementing RBAC and Network Policies
 - Configuring HPA and Cluster Autoscaler
 - Using readiness and liveness probes
 - Applying GitOps or Helm-based deployments
 - Monitoring with Prometheus, Grafana, and Azure Monitor
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Q4. What are the most common Kubernetes issues you have troubleshot?

Answer:

Common issues include:

- Pods stuck in Pending due to insufficient node resources
- ImagePullBackOff due to ACR or secret misconfiguration
- CrashLoopBackOff caused by application startup failures
- PVC binding failures
- Ingress routing or TLS certificate issues

I typically debug using `kubectl describe`, `kubectl logs`, and `events`.

Q5. How do you perform zero-downtime deployments in Kubernetes?**Answer:**

Zero-downtime deployments are achieved using:

- Rolling updates in Deployments
 - Proper readiness and liveness probes
 - Multiple replicas
 - Traffic management via Services or Ingress
 - Canary or blue-green deployments using Helm or Azure DevOps pipelines
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3. Azure Kubernetes Service (AKS)**Q6. What is AKS and how is it different from self-managed Kubernetes?****Answer:**

AKS is a managed Kubernetes service where Azure manages the control plane.

Key differences:

- No control-plane management overhead
 - Built-in Azure AD integration
 - Easier upgrades and scaling
 - Native Azure Monitor and networking integration
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Q7. How do you design AKS for high availability?**Answer:**

High availability in AKS is achieved by:

- Using multiple node pools
 - Enabling Availability Zones
 - Configuring pod anti-affinity
 - Running multiple replicas
 - Using Azure Load Balancer or Application Gateway Ingress Controller
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Q8. How do you upgrade AKS clusters safely?**Answer:**

- Upgrade control plane first
 - Upgrade node pools one at a time
 - Use surge upgrade strategy
 - Validate workloads after upgrade
 - Roll back if required
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4. Azure DevOps – CI/CD**Q9. How do you design CI/CD pipelines using Azure DevOps for AKS?****Answer:**

A typical pipeline includes:

1. Code checkout
2. Build and unit testing

3. Docker image creation
4. Push image to Azure Container Registry (ACR)
5. Deploy to AKS using Helm or kubectl
6. Post-deployment validation

I use YAML-based pipelines for version control and reusability.

Q10. How do you manage secrets in Azure DevOps pipelines?

Answer:

- Azure DevOps variable groups
- Azure Key Vault integration
- Kubernetes Secrets
- Secret injection via CSI driver

Secrets are never hardcoded in pipelines or YAML files.

5. Infrastructure as Code (IaC)

Q11. What IaC tools have you used and how?

Answer:

I have used:

- Terraform for Azure infrastructure provisioning
- ARM/Bicep templates
- Helm for Kubernetes resource templating

IaC helps maintain consistency, repeatability, and faster environment provisioning.

Q12. How do you manage multiple environments using IaC?

Answer:

- Separate state files or workspaces
 - Environment-specific variable files
 - Reusable modules
 - Parameterized Helm charts
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6. Monitoring, Optimization & Troubleshooting

Q13. How do you monitor AKS clusters?

Answer:

- Azure Monitor & Container Insights
 - Prometheus for metrics
 - Grafana for dashboards
 - Alerts on CPU, memory, pod restarts, and node health
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Q14. How do you optimize AKS costs and performance?

Answer:

- Right-size node pools
- Use HPA and Cluster Autoscaler
- Spot node pools for non-prod workloads

- Remove unused resources
 - Optimize container resource requests and limits
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7. Agile & Collaboration

Q15. How do you work in Agile environments?

Answer:

I actively participate in:

- Sprint planning
- Daily standups
- Sprint reviews and retrospectives

I collaborate closely with developers, QA, and product teams to ensure smooth CI/CD and faster releases.

Q16. How do you handle communication with cross-functional teams?

Answer:

- Clear documentation
- Regular sync meetings
- Explaining technical issues in simple terms
- Proactively highlighting risks and improvements

Good communication is critical for DevOps success.

8. Scenario-Based Questions

Q17. A production pod is crashing continuously. How do you debug?

Answer:

1. Check pod status and events
 2. Review application logs
 3. Validate environment variables and secrets
 4. Check resource limits
 5. Roll back if required
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Q18. Pipeline deployment to AKS failed. What steps do you take?

Answer:

- Check pipeline logs
- Validate Docker image availability
- Verify Kubernetes context and permissions
- Check Helm release status
- Fix and redeploy